Ontology Project
Event & Market Temporal Knowledge Graph for NFL

Discipline: Ontology / Knowledge Graph

1. Goal

Design and implement an event-centric temporal knowledge graph (KG) for one chosen **DOMAIN** such as the NFL. The system should:

- Unify games/events, entities (teams, players, actors), markets & odds, and news/textual signals into one coherent schema.
- Encode time correctly: `valid_from`, `valid_to`, timestamps, and sequences of events.
- Support 10–15 non-trivial queries resembling Astera's Sharpe AI needs, fair value engines, and surface constructions.

2. Context: Why This Matters for Astera

AsteraOS's Data & World Model layers (1–3) require a robust ontology and event-centric KG capable of representing:

- Games, plays, injuries, odds movements, market exposures, and resolutions.
- Cross-domain extensibility (sports → macro, politics, weather, crypto events).
- Clean integration with the **Astera Event & Market Ontology Platform** and the **Astera-Code ID space**.

This project represents a candidate-sized slice: one domain, fully wired, but designed with cross-domain generalization in mind.

3. Inputs & Data

You may select any publicly available data for your chosen domain.

3.1 Sports Example (NFL, NBA, Soccer)

- Game schedules, final outcomes

- Team & player rosters
- Box scores or simplified play-by-play
- At least one odds/markets dataset (sportsbook, exchange, prediction market)
- A small news dataset (scraped headlines, RSS feeds, manually curated)

3.2 Non-Sports Example (Elections, Macro, Crypto, Weather)

- Event dates (elections, economic releases, storms, halving events)
- Actors/entities (candidates, central banks, regions)
- Market or contract descriptions + price history (Kalshi, Polymarket-style)
- Associated news/text streams

Simulated data (e.g., a CSV of news headlines) may be incorporated if necessary; the priority is schema design and end-to-end structure.

4. Requirements & Tasks

4.1 A. Ontology Specification (Domain + Cross-Domain)

Design a written ontology including:

Core Entities (Sport-Agnostic Abstractions)

- **Event**: Game, Match, Play, Drive
- **Participant**: Team, Player, Actor
- **Market**, **Contract**, **BettingLine**, **PredictionMarket**
- **OddsMovementEvent**, **MarketResolutionEvent**
- **NewsItem**, **InjuryEvent**
- **Season**, **Competition**, **Tournament**

Relationships & Cardinalities

- Event –[`hasParticipant`]→ Participant

- Event –[hasMarket]→ Market
- Market –[quotedOn]→ Venue (sportsbook/exchange)
- Market –[hasOddsMove]→ OddsMovementEvent
- NewsItem –[refersTo]→ Event / Participant / Market

Temporal Modeling

- valid_from / valid_to for relationships such as playsFor, memberOf, runsFor.
- Timestamps for all event types: Event.start_time, OddsMovementEvent.at_time, etc.

Cross-Domain Abstraction

Provide a short section explaining how the schema extends from your chosen domain to at least two others (e.g., "NFL → NBA" or "NFL → Elections").

Deliverable

- ontology_spec.md with ER-style written descriptions.
- Optional diagrams (PNG/PDF).

4.2 B. Knowledge Graph Implementation

Choose a graph store: Neo4j, Postgres + Apache AGE, RDF triple store, or equivalent.

Data Ingestion Requirements

- Ingest at least 1–2 seasons (or equivalent) of domain events.
- Entities (teams, players, actors) with temporal history.
- Markets & odds (time-series or open/close snapshots).
- News items linked to events/participants/markets.

Load Scripts Must:

- Normalize raw data into ontology schema.
- Write nodes & relationships.

- Be idempotent and support rapid reloading after schema changes.

4.3 C. Query Layer & Example Workflows

Implement 10–15 domain-relevant graph queries, such as:

- Recent injury events affecting a team's implied probabilities.
- Odds movement explanations based on nearby NewsItems.
- Market surfaces aggregated by team, timeframe, or event type.
- Temporal sequences: drives $\rightarrow$ plays $\rightarrow$ market reactions.

These must resemble actual Sharpe AI / Verum engine workflows.

4.4 D. Optional (But Strong): Link Prediction / Completeness Check

Implement a lightweight ML or heuristic-based detector:

- Identify missing Event–Market links (text similarity, embedding threshold).
- Detect anomalous odds moves with no correlated NewsItem.

This demonstrates interaction between KG structure and ML modeling.

5. Deliverables

- **ontology_spec.md**: Full ontology specification.
- **schema_diagram.png/pdf**: Optional but encouraged.
- **Data ingestion scripts/notebooks**: End-to-end loading pipeline.
- **Query examples**: Outputs in JSON or screenshot form.
- **README**: How to run, what to inspect first.

6. Evaluation Criteria

We evaluate based on:

- **Conceptual Clarity**: Clean abstractions, event-centric schema, correct temporal modeling.
- **Generalization**: Ability to extend schema across domains.

- **Practicality**: Queries that support Sharpe AI, fair value engines, & surfaces.
- **Code Quality**: Reproducibility, readability, documentation, idempotent loading.